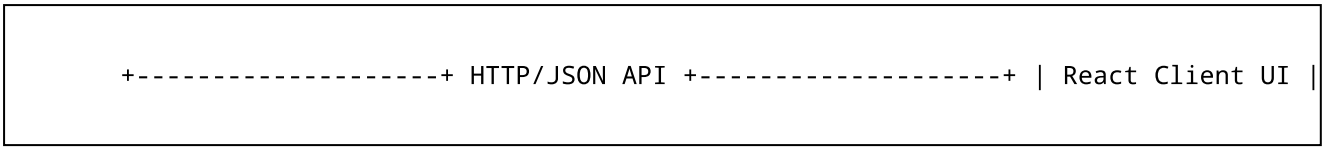
STAFFSYNC PRO

DOCUMENTATION: PHASE 2 — REQUIREMENT ANALYSIS & TECH STACK

Project Name: Nehal_StaffSync | Platform: MERN Architecture

1. DATA FLOW ARCHITECTURE (LEVEL 0 & LEVEL 1)

The system's operational boundaries and data flow pipeline are structured to guarantee stateless communication and strict data validation across tiers.



2. SYSTEM FLOW SPECIFICATIONS

- **User Authentication Flow:** Client transmits credentials via HTTPS POST request to `/api/auth/login`. The Express server verifies credentials against BCrypt hashes stored in MongoDB and returns a signed JWT token upon success.
- **Directory Management Flow:** Authenticated requests containing the Bearer token access `/api/employees`. The controller verifies token validity and returns or modifies employee records accordingly.

3. FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

Functional Requirements (FR)

ID	REQUIREMENT NAME	DETAILED OPERATIONAL DESCRIPTION
FR-01	User Registration	System allows registration of administrative and staff users with password validation.
FR-02	Access Control	System enforces role-based view permissions (Admin vs General Staff).
FR-03	Directory CRUD	Authorized admins can Create, Read, Update, and Delete employee profiles.

Non-Functional Requirements (NFR)

ID	CATEGORY	SPECIFICATION STANDARD
NFR-01	Security	All passwords must be hashed using BCrypt (salt factor 10); API secured via JWT.
NFR-02	Performance	API response latency must remain under 200ms for standard CRUD operations.
NFR-03	Reliability	System state must persist across sessions with automatic MongoDB reconnection.

4. TECHNOLOGY STACK SPECIFICATION

LAYER	TECHNOLOGY	ROLE IN ARCHITECTURE
Frontend Tier	React.js & Tailwind CSS	Single Page Application (SPA) rendering and component styling.
Backend Tier	Node.js & Express.js	RESTful API routing, controller logic, and middleware processing.
Database Tier	MongoDB & Mongoose ODM	Document storage, schema validation, and database querying.